

Group 25

Diabetic insole

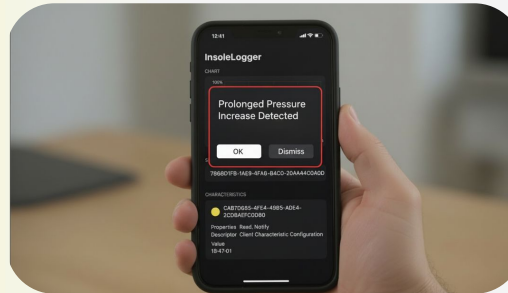
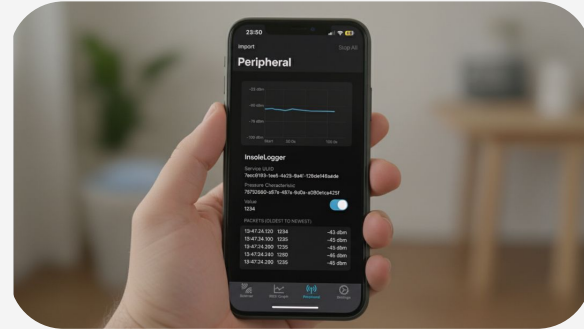
The story

- 2 people have personal relations to people with Type 1 diabetes
- We spoke to both people, both different ages to discuss what challenges they face and what kind of product they could benefit from.
- Identified beeping concerns, pump was loud and disturbing, wanted to manage the condition discreetly
- Looked at multiple solutions for this and the insole was the best solution.
- An estimated 5.8 million people in the UK are living with diabetes.



The solution

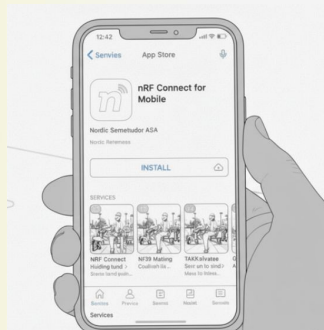
- The InsoleLogger Smart pressure sensing insole
- Measures pressure increase from a swelling foot
- Data is sent to an app which plots readings and notifies user
- of prolonged pressure increase
- Built in SD card to store data which can be saved for long term review.



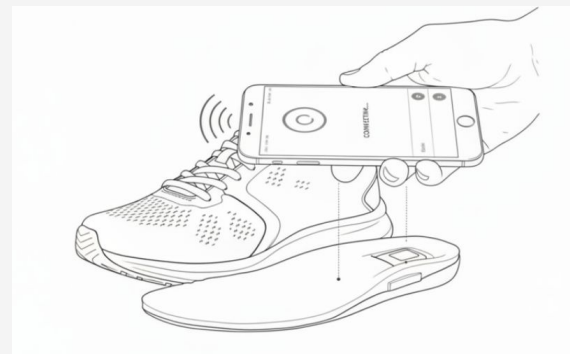
Storyboard



1 - Insert SD Card



2 - Install "nRF Connect"



3 - Connect to InsoleLogger via "Scanner" section

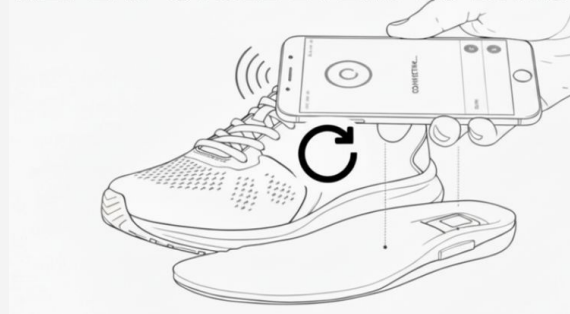


4 - Put insole in shoe



5 - Select "Peripheral" section for pressure readings.

REPEAT CYCLE EVERY 25 DAYS

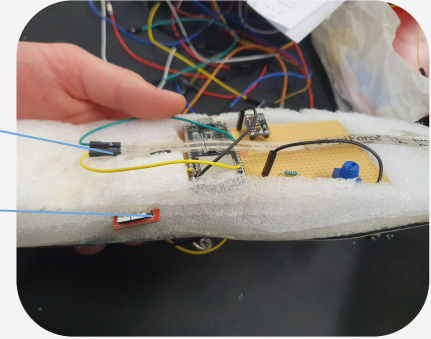
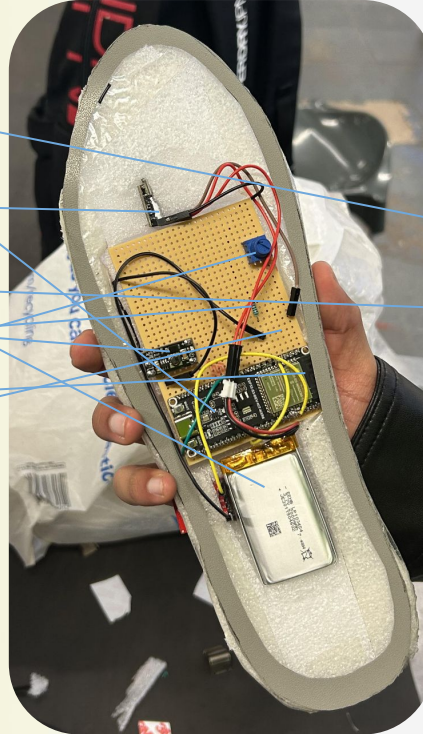


6 - Wear/use as you wish! Battery life is expected ~25 days.

Details

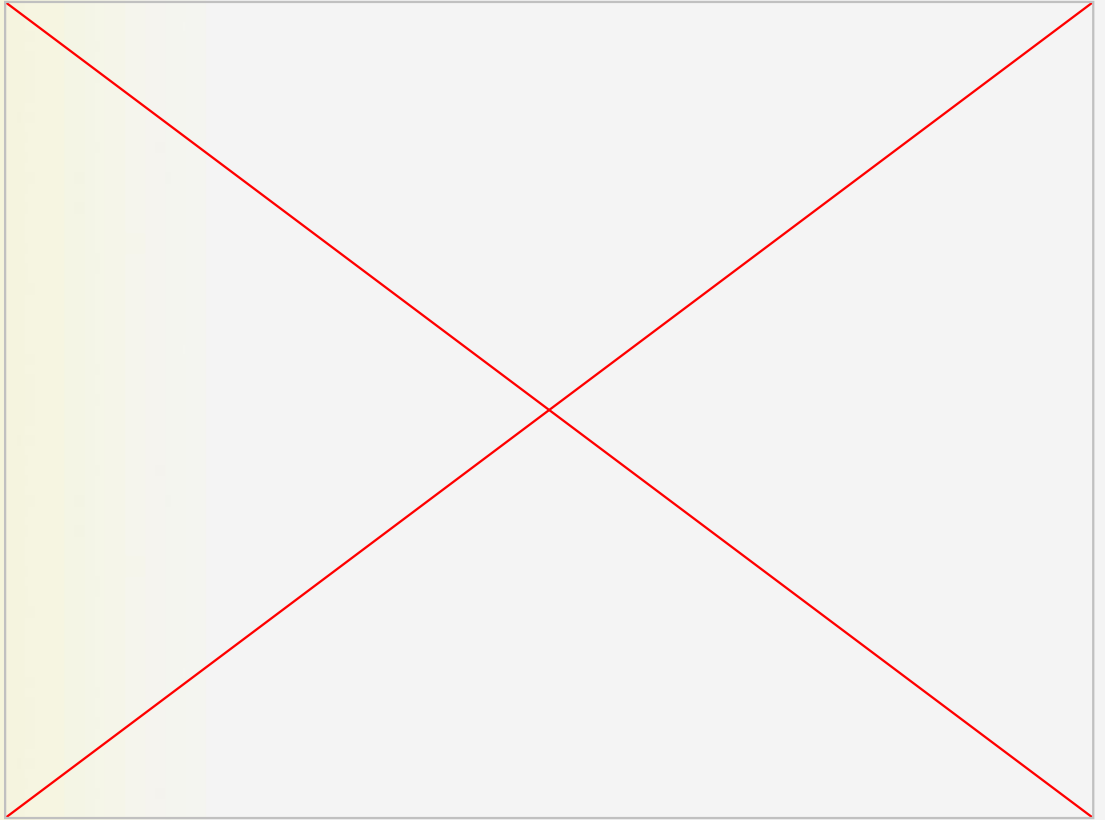
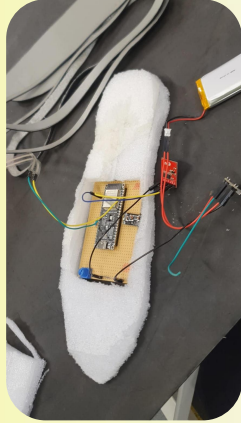
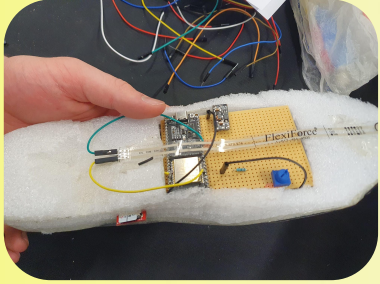
- Waveshare ESP32-S3 development board
- FlexiForce pressure sensor
- Li-Po battery 3.7 V (2000 mAh) with JST connector
- TP4056 charger module
- SD card module (3.3 V compatible) and microSD card
- Small 3.3 V voltage regulator
- Resistor, 10 k Ω
- Jumper cables for connections
- Potentiometer
- Small perfboard

Cotton Exterior lining



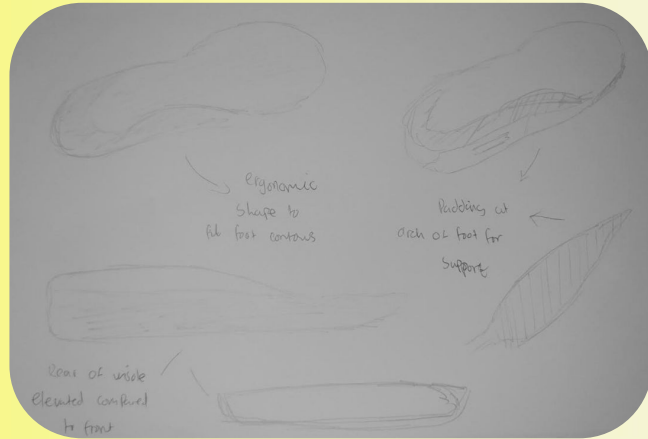
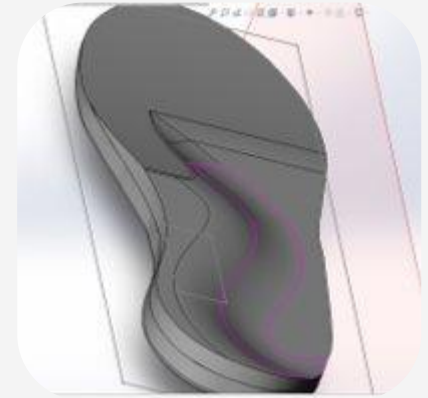
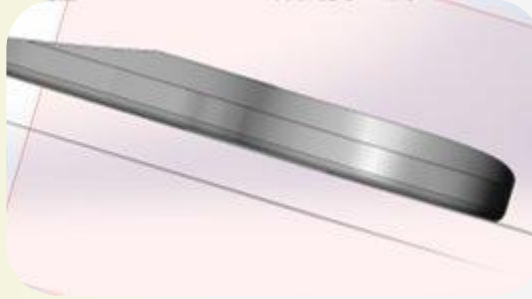
Liquid Silicone Rubber (LSR)

Proof



Design

- Comfort + Ergonomics
- Sizing
- 3D Shape
- Internal Components
- Maximum Sizing



Ergonomic Research

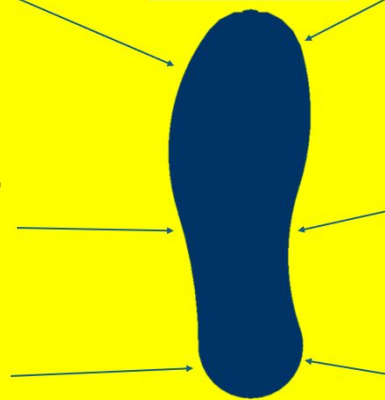
Being a wearable product, the comfort and ergonomics are very important, in particular regards to the feet.

The front toe box is the thinnest part of the insole, this is to allow for flexibility in activities such as running or jumping.



The front of the insole has a teardrop shape which is specifically shaped for either left or right foot, the area is very flat and thin, to allow for movement and flexibility of the toes

The typical height of the arch is 60% of the length of the insole, however this can vary as some insoles are designed to provide more support, for example to those with flat feet.



Possibly the most important part of the insole is the mid section arch support, this is a raised gradual curve which sits fitted into the bridge of the foot, this arch is important to shift weight while walking and reduce stress on the midfoot area.

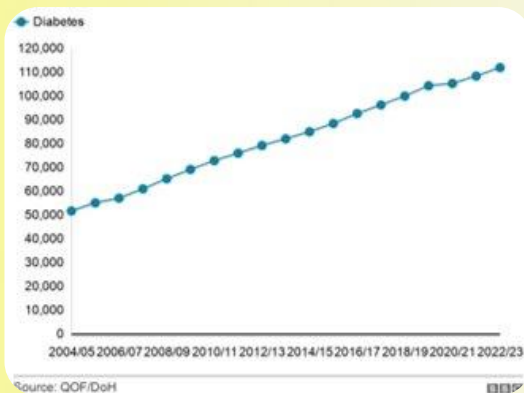


The heel section is raised and flares outward to absorb impact pressure and reduce the shock impact up to the shins and knees, the shape also helps to prevent heel roll ins.

The heel cup has a concave shape and plays outwards in a bowl shape to match the shape of the ball of the feet, it gradually inclines towards a peak of around 10-14 mm above base height

Market research

- Diabetes is a growing issue in the UK
- Outpatient care spending has doubled over last decade to £1.92 billion
- NHS: £10.7 Billion in 2025 to £18 billion in 2035
- £825 Million on Outpatient care from NHS



x
Age – 20
Occupation – Student



Has been suffering from diabetes since very young and wants an effective, affordable and discreet way of managing their condition. - is a university student so a significant pain point of managing the condition is the excessive price involved, some devices designed to measure insulin spikes require being replaced in as little time as every 10 days, and with substantial prices involved of up to £60 per unit, costs add up and greatly affects the accessibility of care, as well as this, being a university student, - finds the current methods of insulin warnings very intrusive with risk of the warning disturbing a lecture, so would prefer something more understated.

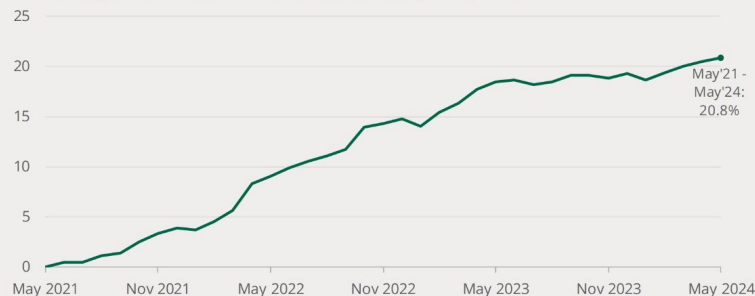
x
Age - 42
Occupation – Bank manager



Has been suffering with diabetes since they were 16 years old, - works as a manager of a bank and is constantly in meetings or communicating with customers. The current methods of warning for an insulin spike are too invasive and apparent, with loud beeping reminders which can seem unprofessional or disturbing in a professional setting, leaving – feeling embarrassed, however – cannot turn off this warning as it is important to know when insulin is spiked so it can be dealt with.

UK consumer prices increased by over 20% in three years to May 2024

% change in price level since May 2021 on CPI measure (up to May 2024)



Source: ONS, CPI price level, series D7BT (June 2024 update)

Ethical considerations

- Material Choice – Toxicity, breathability
- Data Security
- Mobility Interference + Ease of Install
- Regulatory Compliance
- Build Quality – Finishings and Electronics Placement



Trends

Moving forward:

- Test on patients with actual condition
- Clinical trials
- Ask for feedback to develop product.
- Improvements

Current products:

- Only one privately distributed product exists.
- Images shown on right are attachable, not independent products.
- Our product identifies a gap in the market and offers a valuable solution.



£105



£105



Financial

- Li-Po battery- £9
- TP4056 module- £3
- Voltage regulator- £2
- Potentiometer- £1
- Resistors, perfboard, jumper cables ~£2
- ESP32-S3- £10.60
- Flexiforce pressure sensor- £21.68
- SD Card reader- £4.31
- SD Card- £2.40

Totals £55.99,

+ £5 materials cost, = £60.99

Selling price: £74.99

Profit = £14.99 per unit, more with mass manufacturing
£218.85 profit per person per year.

- G6/G7 sensors, £1,850 a year if they actually last every 10 days, £3,650 if they fall off every 5 days.
- Our product would cost £1,094.85 a year



£105



£105



December
2025

Questions?